METRIC

MIL-I-24772(SH)
17 March 1992

MILITARY SPECIFICATION

INDICATOR, OXYGEN, COMBUSTIBLE GAS, CARBON MONOXIDE, AND HYDROGEN SULFIDE, PORTABLE (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for portable indicators used to detect and indicate existing levels of oxygen, combustible gas, hydrogen sulfide, and carbon monoxide. The indicator can also evaluate the time weighted average of hydrogen sulfide and carbon monoxide exposure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

DISTIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:03Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisional Items (Repair Parts): Packaging of.

MIL-F-24385 - Fire Extinguishing Agent, Aqueous Film-Forming Foam, (AFFF) Liquid Concentrate, for Fresh and Seawater.

MIL-B-55130 - Batteries, Rechargeable, Sealed Nickel Cadmium

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-462 - Electromagnetic Interference Characteristics,-Measurement of.

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

MIL-STD-831 - Test Reports, Preparation of.

MIL-STD-1399 - Interface Standard for Shipboard Systems, Section 300, Electric Power, Alternating Current (Metric).

MIL-STD-2036 - General Requirement For Electronic Equipment Specifications

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, and publications. The following other Government documents, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)
CFR 29 - LABOR

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those cited in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
National Electrical Code - Manual Number 70

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, Massachusetts 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3 in accordance with 4.3).

3.2 Selection of material and parts. Material and parts shall meet UL certification in accordance with 29 CFR 1926. The finished product shall conform to all requirements of this document and shall perform their intended function.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

3.3 Construction. This indicator shall provide digital readouts for monitoring oxygen, the lower explosive limit, hydrogen sulfide and carbon monoxide. The indicator shall be designed to have characteristics, features and accessories in accordance with 3.3.1 and 3.3.2.

3.3.1 Features.

3.3.1.1 Electrical circuits. Solid state circuits shall be employed and may be microprocessor controlled.

3.3.1.2 Switches and controls. The switches and controls shall be of the sealed membrane type or be recessed in the housing.

3.3.1.3 Liquid crystal display. A 3-1/2 digit display minimum shall be used to display the sensed atmosphere levels. The minimum digit height shall be 9.52 millimeters.

3.3.1.4 Display backlight. The indicator shall have a display light to enhance readability in dark environments.The backlight shall be of sufficient brilliance to make the display readable in all degrees of subdued light including total darkness. Backlight shall be controlled by normally open momentary contact switch or photodiode or similar sensor.

3.3.1.5 Low battery level indicator. A low battery level indicator shall be activated on the liquid crystal display to indicate battery charge needed.

3.3.1.6 Self testing circuitry. The indicator shall have a self testing circuit to diagnose sensor, integrated circuit, or wiring damage or failure and shall self test upon power-up. Self test circuitry shall not test or adjust calibration.

3.3.1.7 Alarm. An audible (80 + 5 decibel (dB) at 0.3 meter distance) and visual alarm displayed for the cause of the alarm shall be provided.

3.3.1.8 Power source. The power supply shall consist of a leak-proof, rechargeable sealed nickel cadmium storage battery conforming to MIL-B-55130. The nominal voltage rating of the battery shall be 6 volts direct current (dc) or less. The capacity shall provide power to operate the indicator for a minimum of 9 hours from a full charge condition. The battery shall be fully recharged in 8 or less hours after 9 hours of continuous operation.

3.3.1.9 Calibration and span. Calibration circuitry or adjustment devices, such as potentiometers, necessary for zero and span adjustments by the user shall be incorporated, in easily accessible locations.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:03Z
Check the source to verify that this is the currentversion before use.

3.3.1.10 Repairability. The unit shall have modular construction to enable repair by module replacement. The indicator shall be designed so that removal or replacement of any module shall not be detrimental to the indicator.

3.3.1.11 Maximum lock. The indicator shall retain a lock on the highest lower explosive limit (LEL), hydrogen sulfide, carbon monoxide reading, and the lowest oxygen reading.

3.3.1.12 Ranges and alarm set points. Indicator ranges shall be as follows:

(a) Oxygen 0-25 percent.
(b) Combustible 0-100 percent LEL.
(c) Hydrogen sulfide 0-500 parts per million (ppm).
(d) Carbon monoxide 0-999 ppm.

Alarm ranges shall be as follows:

(a) Oxygen 19.5 - 23.0 percent (both high and low alarms).
(b) Combustible 10 percent LEL.
(c) Hydrogen sulfide 50 ppm instantaneous.
10 ppm time weighted average.
15 ppm short term exposure limit.
(d) Carbon monoxide 200 ppm instantaneous.
35 ppm time weighted average.
100 ppm short term exposure limit.

3.3.1.13 Time weighted average. The indicator shall be capable of calculating the time weighted average (TWA) or accumulated exposure for toxic levels since the last power on and digitally display the accumulated exposure level or accumulated exposure of hydrogen sulfide and carbon monoxide. The indicator shall be capable of retaining the calculated TWA if the indicator is turned off.

Ranges for TWA shall be as follows:

(a) Hydrogen sulfide 0-500 ppm.
(b) Carbon monoxide 0-999 ppm.

3.3.1.14 Elapsed time indicator. The indicator shall display the elapsed time in hour-minute format from the onset of TWA start.

3.3.1.15 Size. The size of the indicator shall be not greater than 26.0 by 13.0 by 6.5 centimeters (cm).

3.3.2 Accessories. The following accessories shall be provided with the indicator (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:03Z
Check the source to verify that this is the current version before use.

3.3.2.1 Battery charger. A battery charging circuit capable of operating from a single phase, 60 hertz (Hz), 105 to 120 volts ac source as specified in section 300 of MIL-STD-1399 shall be provided. The charging unit shall not damage the batteries when the battery is fully charged and the charging circuit is fully energized. A battery charging indicator lamp shall be energized and remain energized during charging of the battery. The indicator shall be operational while the battery charger is in operation. The leakage current of the indicator shall not exceed 5 milliamperes (mA) to ground with the battery charging circuit in operation.

3.3.2.2 Detail of components. Sensors for oxygen, carbon monoxide and hydrogen sulfide shall be of electrochemical design with temperature compensating device. The combustible gas sensor shall be a catalytic type.

3.3.2.3 Remote sampling hose. The indicator shall be supplied with 6.10 meters of non-absorbing remote sampling hose.

3.3.2.4 Remote sampling pumps. A remote sampling pump to draw remote air samples to the indicator shall be provided. The pump may have its own rechargeable power source. The pump flow rate and sampling hose diameter shall provide the response time as specified in 3.4.3. A remote sampling manually operated aspirator bulb shall also be provided and be operable independently of the power driven pump.

3.3.2.5 Jewelers' screwdriver . A jewelers' screwdriver shall be provided to adjust the alarm set points and for calibration procedures.

3.3.2.6 Carrying strap. The indicator shall be provided with a carrying strap that can be shortened or lengthened to hang the indicator in visible range on the wearer's chest. The strap will have the capability of being removed.

3.3.2.7 Earphones. The indicator shall be supplied with earphones for operating in noisy environments.

3.3.2.8 Calibration kit. A calibration kit to ensure accurate response to known atmospheres shall be provided. The kit shall include one bottle each of methane, carbon monoxide and hydrogen sulfide; a gas flow regulator; appropriate fittings and hoses; and a carrying case. The gas flow regulator rate shall be determined by the manufacturer for optimum flow rate for calibration of the instrument. Enough calibration gas shall be supplied to provide in excess of 50 calibrations, in the following concentrations:

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:03Z
Check the source to verify that this is the current version before use.

(a) Methane - 2.5 percent by volume plus 2 percent.
(b) Hydrogen sulfide - 75 ppm plus 5 percent.
(c) Carbon monoxide - 500 ppm plus 2 percent.

3.3.2.9 Carrying case. A plastic carrying case to hold and protect the remote sampling pump, battery charger, any necessary tools and the indicator shall be provided.

3.4 Performance.

3.4.1 Calibration. The indicator shall be calibrated by use of calibration potentiometers for zero and span adjustment which are easily accessible although sealed when in operation. Calibration kit gas quality and regulator flow rate shall meet the requirements of 3.3.2.8. Testing shall be in accordance with 4.6.1.

3.4.2 Stability. During continuous operation at any oxygen, combustible gas, hydrogen sulfide, or carbon monoxide level for a minimum period of 15 minutes, the meter indication shall be stable within plus 0.1 percent oxygen, plus 1.0 percent LEL, and plus 1 ppm hydrogen sulfide and 5 ppm carbon monoxide. Testing shall be in accordance with 4.6.2.

3.4.3 Response time. The indicator shall provide a meter indication for 90 percent of final value either rising or falling as follows:

(a) Oxygen, within 10 seconds.
(b) Combustible gas (methane), within 30 seconds.
(c) Hydrogen sulfide, within 90 seconds (180 seconds falling).
(d) Carbon monoxide, within 60 seconds.
(e) Combustible gas (methane) with power driven pump and 6.10 meters of remote sampling hose, within 90 seconds (60 seconds draw time).

Testing shall be in accordance with 4.6.3.

3.4.4 Accuracy. The indicator shall be accurate in accordance with the following requirements:

(a) Oxygen plus 0.5 percent from 0-25 percent oxygen by volume.
(b) Combustible, plus 5.0 percent LEL from 0-100 percent LEL.
(c) Hydrogen sulfide, plus 5.0 ppm from 0-100 ppm.
(d) Carbon monoxide, plus 5.0 ppm from 0-250 ppm.

Testing shall be in accordance with 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:03Z
Check the source to verify that this is the current version before use.

3.4.5 Repeatability. The indicator shall have repeatable meter indications and alarm setpoints as follows:

(a) Oxygen, plus 0.2 percent oxygen meter indication, plus 0.1 percent oxygen alarm setpoint.
(b) Combustible gas, plus 2.0 percent LEL meter indication plus 1.0 percent LEL alarm setpoint.
(c) Hydrogen sulfide, plus 1.0 ppm meter indication, plus 1.0 ppm alarm setpoint.
(d) Carbon monoxide, plus 2.0 ppm meter indication, plus 1.0 ppm alarm setpoint.

Testing shall be in accordance with 4.6.5

3.4.6 Inclination. Performance of the indicator, remote sampling pump, and battery charger shall not be affected by inclination to any attitude. Testing shall be as specified in 4.6.6.

3.4.7 Temperature. Performance of the indicator, sampling pump, and battery charger shall not be adversely affected by operation over the temperature range of 0 to plus 40 degrees Celsius (°C) or by storage over the temperature range of minus 20 to plus 60°C. Testing shall be in accordance with 4.6.7.

3.4.8 Humidity. Performance of the indicator, sampling pump, and battery charger shall not be affected by the humidity test as specified in 4.6.8.

3.4.9 Vibration. Performance of the indicator, sampling pump, and battery charger shall not be affected by the vibration test as specified in 4.6.9. There shall be no evidence of mechanical damage or loosening of parts on the indicator or accessories following the vibration test.

3.4.10 Salt fog. Performance of the indicator, sampling pump, and battery charger shall not be affected by the salt fog test as specified in 4.6.10. There shall be no evidence of rust or corrosion on the indicator or accessories.

3.4.11 Transit drop. Performance of the indicator, sampling pump, and battery charger shall not be affected by the transit drop test as specified in 4.6.11. There shall be no evidence of mechanical damage or loosening of parts on the indicator or accessories following the transit drop test.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:03Z
Check the source to verify that this is the current version before use.

3.4.12 Electromagnetic interference. Performance of the indicator, sampling pump, and battery charger shall not be affected by the electro-magnetic test as specified in 4.6.12.

3.4.13 Flashback protection. The chamber housing the filament of the combustible gas indicator shall be provided with flame arrestors in order to prevent flashbacks from igniting outside gases. The indicator shall pass the test specified in 4.6.13 without ignition of the gases. The indicator, remote sampling pump, and battery chargers shall be intrinsically safe for operation in class I, division I, (article 500-4), group C and D (article 500-2) environments as defined by manual 70 of the National Electrical Code.

3.4.14 Electrical. Performance of the indicator, sampling pump, and battery charger shall not be affected by the steady state voltage and frequency, transient voltage, transient frequency, spike voltage, and power interruption tests as specified in 4.6.14.

3.5 Life.

3.5.1 Sensors. Sensors shall have a minimum of 1-year warranty period in which sensors shall be replaced if defective. Self stamp addressed cards shall be provided with all sensors and the cards, upon removal of the sensor from its packaging, shall be filled out and returned to the contractor for an accurate first day of service. If a sensor is returned within the 1-year warranty period it shall be replaced free of charge if found to be defective, and mailed to the users' designated address. Each individual sensor must have a unique serial number, in order to provide traceability and proof of the warranty claim.

3.5.2 Unit. The unit shall remain operational in the nondefective modes if one or more sensors or filament heads are not operating. A warning on the LCD indicator shall inform the operator of the defective sensor or filament. Unit life shall be tested in accordance with 4.6.15.

3.5.3 Battery. The battery shall provide the indicator with 9 continuous hours of operation at ambient room temperature. Testing shall be in accordance with 4.6.15.

3.6 Drawings. (See 6.3 and Appendix)

3.7 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award for the calibration gases containing methane, carbon monoxide, or hydrogen sulphide. The MSDS shall be included with each shipment of the material covered by this specification (see 6.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

3.8 Operating instructions. Brief essential operating instructions shall be permanently and legibly mounted on the indicator case for ready reference.

3.9 Workmanship. The workmanship on the indicators shall be free from imperfections or defects of manufacture which may affect their appearance, serviceability, or operation. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; marking of parts and accessories; thoroughness of soldering and painting; alignment of parts and tightness of assembly screws, bolts, and so forth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operation, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations and tests as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

TABLE I. First article examinations and tests.

Examinations and tests	Requirements Paragraph	Test Paragraph
General examination	3.2 to 3.3	4.5.1
Operation and performance		
Calibration	3.4.1	4.6.1
Stability	3.4.2	4.6.2
Response	3.4.3	4.6.3
Accuracy	3.4.4	4.6.4
Repeatability	3.4.5	4.6.5
Inclination	3.4.6	4.6.6
Temperature	3.4.7	4.6.7
Humidity	3.4.8	4.6.8
Vibration	3.4.9	4.6.9
Salt fog	3.4.10	4.6.10
Transit drop	3.4.11	4.6.11
Electromagnetic interference	3.4.12	4.6.12
Flashback	3.4.13	4.6.13
Electrical	3.4.14	4.6.14
Life	3.5	4.6.15

4.3.1 Test conditions. Unless otherwise specified (see 6.2), all tests required by this specification shall be conducted at standard atmospheric pressure, a temperature of 25 ± 5°C, and a relative humidity of 80 percent or less.

4.3.2 First article sample. Two indicator assemblies, as specified in 6.5, shall be tested as specified in table I. Together with a test sample, the manufacturer shall submit one reproducible copy of the drawings and two printed copies of instructions with illustrations and diagrams, covering the operation of the indicator.

4.3.3 Tests. Indicators used for first article inspection shall be subjected to the tests shown in table I in the order listed. Failure of an indicator to comply with any of the requirements listed shall be cause for rejection of the first article and require that corrective action be implemented by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on every lot of indicators acquired under this specification. Quality conformance inspection shall consist of the following groups and tests:

(a) Group A (see 4.4.2.1).
(b) Group B (see 4.4.2.2).
(c) Group C (see 4.4.2.3).

4.4.1 Inspection lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as a all the indicators of the same type, style, and size produced in one facility using the same production processes and materials and offered for delivery at the same time.

4.4.2 Sampling for quality conformance inspections. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed indicators in accordance with table II, table III, and paragraph 4.4.2.3 and inspect them in accordance with 4.4.3. If one or more defects are found in any sample, the enitre lot shall be rejected. The contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot whic shall be inspected in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion records of inspections, tests, and any resulting rejections.

Table II. Sample size for group A examinations.

Lot Size	Sample Size
2 to 8 9 to 90 91 to 150	All 8 12
151 to 280 281 to 500 501 to 1,200	19 21 27
1,201 to 3,200 3,201 to 10,000 10,001 to 35,000	35 38 46

4.4.2.2 Sampling for group B tests. A sample of indicators shall be selected from each lot in accordance with table III and subjected to each of the group B examinations and tests specified in 4.4.3, in the order shown. Group B tests shall be performed on sample units that have passed the group A tests.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:03Z
Check the source to verify that this is the currentversion before use.

Table III. Sample size for group B tests.

Lot Size	Sample size
up to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1,200	15
1,201 to 3,200	18
3,201 to 10,000	22

4.4.2.3 Sampling for group C inspection. Every 3 years, two indicators of each class and type produced during the period shall be subjected to the group C inspections specified in 4.4.3. The indicators selected shall be sample units that have passed the group A and group B tests. Group C tests shall be required when the basic design of the indicators or the process or material has been changed.

4.4.3 Examination and test. The groups A, B, and C examination and tests of table IV shall be performed in the order listed.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:03Z
Check the source to verify that this is the currenversion before use.

TABLE IV. Quality conformance inspection.

Examination or test	Requirement paragraph	Test paragraph
Group A General examination	3.2, 3.3	4.5.1
Group B Calibration Stability Response Accuracy	 3.4.1 3.4.2 3.4.3 3.4.4	 4.6.1 4.6.2 4.6.3 4.6.4
Group C Repeatability Inclination Temperature Humidity Vibration Salt fog Transit drop Electromagnetic interference Flashback Electrical Life	 3.4.5 3.4.6 3.4.7 3.4.8 3.4.9 3.4.10 3.4.11 3.4.12 3.4.13 3.4.14 3.5	 4.6.5 4.6.6 4.6.7 4.6.8 4.6.9 4.6.10 4.6.11 4.6.12 4.6.13 4.6.14 4.6.15

4.4.3.1 Quality conformance test report. Quality conformance test reports shall be in accordance with MIL-STD-831.

4.4.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the contracting officer.

4.4.5 Disposition of test samples. Sample units that have been subjected to and passed the group A and group B tests may be delivered on a contract. Group C test samples shall not be delivered on a contract.

4.5 Visual examination and test procedure. The indicator together with all accessories (remote sampling hose, remote sampling pumps, battery charger, and carrying case) shall be referred to as the indicator assembly in section 4.5 and 4.6.

4.5.1 General examination. Each indicator assembly shall be subject to a general examination in accordance with MIL-STD-2036. The examination shall verify compliance with the material and requirements specified in 3.2 and 3.3.

4.6 Operation and performance.

4.6.1 Calibration. Calibration gas quality shall be certified to meet the requirements of 3.3.2.8 and tested for this quality using a gas chromatograph or equivalent instrumentation. Five bottles of each gas shall be tested.

4.6.2 Stability. Power from the indicators fully recharged batteries shall be applied to each indicator under test for a maximum of 10 seconds before it is calibrated. The indicator shall then be operated, without recalibration, for ten 15-minute periods while independently sampling atmospheres containing the following known concentrations:

(a)	Oxygen	- 15, 20 and 25 percent.
(b)	Methane	- 10, 50 and 90 percent LEL.
(c)	Tetraethyl lead gasoline	- 30 percent LEL.
(d)	Hydrogen sulfide	- 10 ppm by volume.
(e)	Carbon monoxide	- 100 ppm by volume.
(f)	Halon 1301	- 7 percent by volume.

For one 15-minute cycle of methane concentration tests, the power driven remote sampling pump and 6.10 meters of remote sampling hose shall be used. The indicator shall meet the stability requirements of 3.4.2. During the stability test special attention shall be taken to record the response time both rising and falling (see 4.6.3), accuracy to known concentrations (see 4.6.4) and repeatability during the ten periods (see 4.6.5).

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:03Z
Check the source to verify that this is the current version before use.

4.6.3 Response time. The time of response measured during the stability test (see 4.6.2) shall be not more than that specified in 3.4.3.

4.6.4 Accuracy. The accuracy measured during the stability test (4.6.2) shall meet the requirements of 3.4.4.

4.6.5 Repeatability. Data obtained during the stability test shall meet the requirements of 3.4.5.

4.6.6 Inclination. The indicator and remote sampling pump operated together and separately shall be inclined 90 degrees to each side respectively of the normal operating position and remain within the accuracy requirements of 3.4.4.

4.6.7 Temperature. A high temperature test shall be conducted for the indicator assembly in accordance with method 501, procedure II of MIL-STD-810. The highest non-operating temperature shall be plus 60°C. The highest operating temperature shall be plus 40°C. A low temperature test shall be conducted in accordance with method 502, procedure I of MIL-STD-810. The lowest non-operating temperature shall be minus 20°C. The lowest operating temperature shall be 0°C. The indicator shall meet the requirements of 3.4.7 and remain within the accuracy requirements of 3.4.4.

4.6.8 Humidity. The humidity test shall be conducted for the indicator assembly in accordance with method 501, procedure II of MIL-STD-810. The highest operating temperature shall be plus 40°C. The indicator shall meet the requirements of 3.4.8 and remain within the accuracy requirements of 3.4.4 during and after the test.

4.6.9 Vibration. The indicator assembly shall be tested in accordance with type I of MIL-STD-167-1. During the test, all operating controls will be observed for a change in status. After the test, the indicator shall be subject to a visual examination for evidence of mechanical damage or loosening of parts. The indicator assembly shall be securely mounted on a vibration table during this test. After testing the indication shall meet the stability, response time, and accuracy requirements of 3.4.2, 3.4.3, and 3.4.4 respectively.

4.6.10 Salt fog. The salt fog test shall be conducted for the indicator assembly in accordance with method 509, procedure I, of MIL-STD-810. The indicator shall not be operated during the test. After the test, the indicator shall be subject to the stability test, 4.6.2, and shall meet the accuracy requirements of 3.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

4.6.11 Transit drop. The indicator assembly, in the carrying case, shall be tested in accordance with method 516.3, procedure I, of MIL-STD-810. The indicator shall meet the accuracy requirements of 3.4.4 after the transit drop test.

4.6.12 Electromagnetic interference. The indicator shall be tested in accordance with MIL-STD-462. During the test the indicator shall meet the accuracy requirements of 3.4.4 and the requirements of CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 of MIL-STD-462.

4.6.13 Flashback. Each indicator under test shall meet the requirements of the explosive atmosphere test as shown in method 511.1, procedure I of MIL-STD-810. During the test the charged battery charging circuit shall be connected and disconnected in the test atmosphere twice.

4.6.14 Electrical. The indicator assembly while connected to the battery charger, shall be subjected to the following electrical tests in accordance with MIL-STD-2036:

(a) Steady state voltage and frequency.
(b) Transient voltage.
(c) Transient frequency.
(d) Spike voltage.
(e) Power interruption.

4.6.15 Life. The indicator and remote sampling pump shall be operated from a fully charged condition for five, 9-hour cycles under ambient conditions. During the test cycles the low battery indicator shall be observed and recorded as a percentage of full battery life at onset. The battery recharge time shall also be limited to 8 hours.

4.7 Inspection of packaging. Sample package and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 Packing and marking. Indicators shall be preserved level A, C, or commercial, packed level A, B, C, or commercial (see 6.2) and marked in accordance with MIL-E-17555. Hazardous materials shall conform to the Department of Transportation regulations as specified in MIL-E-17555.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:03Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicator covered under this specification is intended for use by ship's damage control and gas free engineering personnel to detect and indicate the short-term exposure limits for oxygen, combustible gas, hydrogen sulfide, and carbon monoxide.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Quantity of items required
(d) When sample for first article inspection is required (see 3.1)
(e) Accessories required (see 3.3.2)
(f) Test condition required (see 4.3.1)
(g) Selection of applicable levels of packing and marking (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
3.6	DI-DRPR-80651	Engineering Drawings associated lists	Level II

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:03Z
Check the source to verify that this is the current version before use.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first 3 production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instruction in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.7 Material safety data sheet (MSDS). Contracting officers should identify those activities requiring copies of MSDS's. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when materials and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Subject term (key word) listing.

Analyzer
Gas monitor

Preparing activity:
Navy - SH
(Project 6665-N491)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.

APPENDIX
ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. Scope

10.1 Scope. This appendix covers the technical content requirements that shall be included on drawings. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. Drawings shall contain the following information:

(a) Manufacturer's name and identification number of equipment.

(b) List of descriptive data including:

(1) Complete rating showing battery voltage, battery charging input voltage, and frequency, outputs, and others.
(2) Enclosure size and weight.
(3) Notation of the corrosion resistance of all hardware.
(4) Special features not covered by this specification.

(c) A complete block diagram illustrating circuit operation.

(d) List of repair parts and special tools in table form.

(e) A table or list of parts such as printed circuit boards, potentiometers, Acapacitors, and similar ıtems giving piece number, type, and ratıng of each. Ratıngs shall include both manufacturer's and the specific application rating.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:03Z
Check the source to verify that this is the currenversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-I-24772	2 DOCUMENT DATE (YYMMDD) 17 March 1992

3 DOCUMENT TITLE
Indicator, Oxygen, Combustible Gas, Carbon Monoxide, and Hydrogen Sulfide

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed*)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME Commander Naval Sea Systems Command (SEA 55X22)	b TELEPHONE (Include Area Code) (1) Commercial: (703) 602-0323 (2) AUTOVON: (AV) 332-0323
c ADDRESS (Include Zip Code) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403, Falls Church VA 22041 3466 Telephone (703) 756-2340 AUTOVON 289 2340

DD Form 1426, OCT 89 *Previous editions are obsolete* 198-290

Source: https://assist.dla.mil - Downloaded: 2026-10-20T02:03Z
Check the source to verify that this is the current version before use.